BIOL3402 Cell Biology and Cell Technology

Damian Lau

2025-2026 Semester 1

Introduction

This course begins with a discussion of cell biology, covering the Cell, Membrane, Nuclear Transport, Techniques in Plant Cell Culture, Transport across the Plasma Membrane, Ion Channels and Optogenetics, Junction, Cytoskeleton, Cellular Adhesion molecules, Cell Migration, Cell Signaling and Techniques in Animal Cell Culture. This course aims to introduce the cellular knowledge. Since this is also a Lab-Lecture based course, this course also provide different culturing knowledge and laboratory skills. Upon completion, students will be familiar with the cell biology and cell technology conceptually and practically.

Contents

1 Lecture 1 & 2 The Cell/Membrane & Nuclear Transport

1. Membrane Structure
   - The Lipid Bilayer
   - Membrane Proteins
2. Functions : Selective Barriers (Allowing the inside of cell to be different from the outside)
3. Compartments of a cell are surrounded by Membranes:
   - Endoplasmic Reticulum
   - Nucleus
   - Peroxisome (single layer membrane)
   - Lysosome
   - Golgi
   - Mitochondrion
   - Transport Vesicles
4. Nucleus and Mitochondria (Also Chloroplasts in plant have double membranes

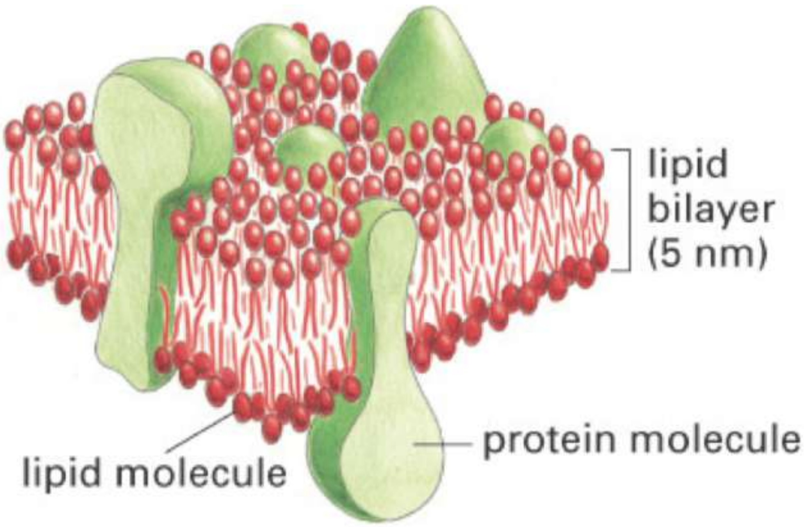

Figure 1: The Lipid Bilayer

5. Membrane lipids are amphipathic molecules
   - A hydrophilic head (Polar) form Hydrogen bonds with water
   - Hydrophobic tails (Non-polar) are excluded by water molecules
   - Phospholipids is the most common lipid in mammalian membranes
6. Three major classes of membrane lipids (All have polar head groups)
   - Phospholipid : Phosphatidylserine (Negative Charged)
   - Sterol : Cholestrol
   - Glycolipid : Galactocerebroside
7. Polar molecules can dissolve in other polar molecules
8. Amphiphilic molecules in an aqueous environment
   - Form micelles or bilayer in water
   - Closed structure : Avoid the exposure of the hydrophobic hydrocarbon tails to water
   - The shape of lipid pack control the structure of phospholipid formation
9. All phospholipid are only synthesized on the cytosolic half of the endoplasmic reticulum membrane

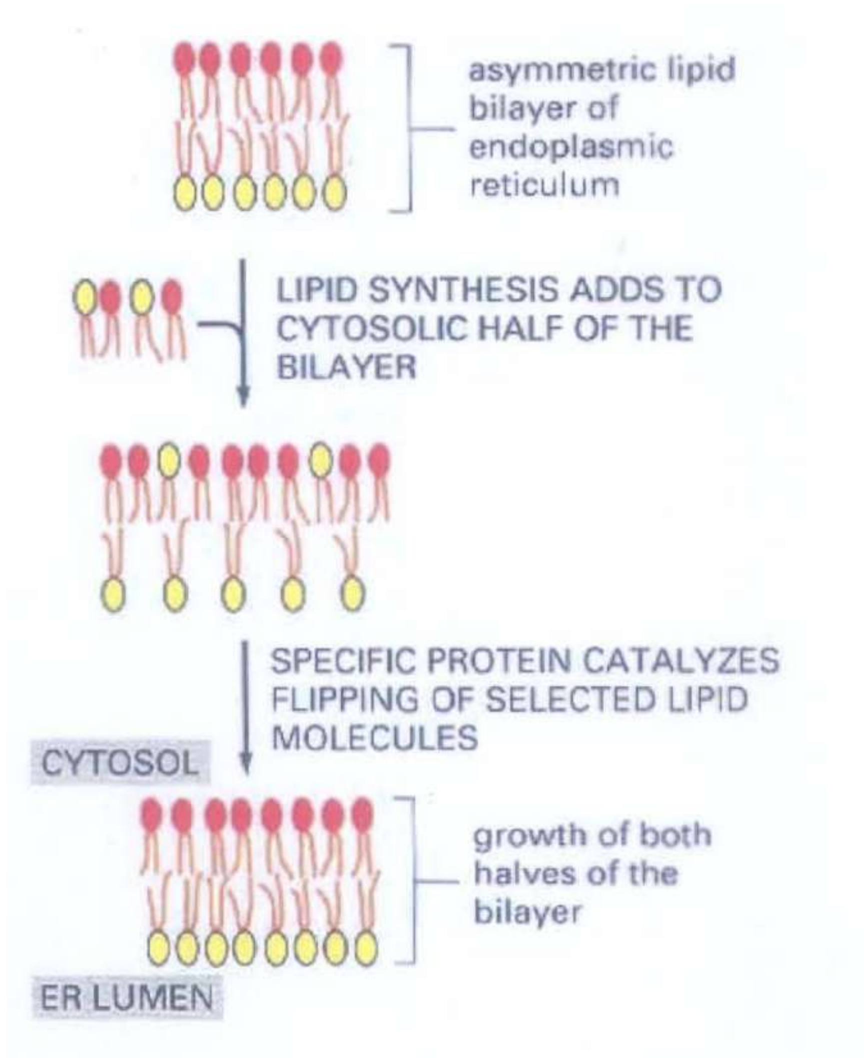

Figure 2: New Lipid Flipping

10. New Lipid Flipping

    - New lipid cannot flip from one monolayer to another one spontaneously (Because moving a Polar head group through the membrane is highly energetically unfavorable
    - It requires Membrane-Bound Phospholipid Translocator Protein (Flippases)
    - Phosphatidylcholine, Sphingomyelin, Glycolipids are in outer leaflet
    - Phosphatidylserine, Phosphatidylinositol, Phosphatidylethanolamine

are in inner leaflet

- Phospholipids and Glycolipid are distributed asymmetrically in plasma membrane lipid bilayer
- Flip-flopping is to maintain the structure of asymmetry and facilitating cell signalling

11. Fluidity of the Membrane
    - Lateral diffusion (Move Left-Right direction) in 2mm/sec
    - Individual phospholipid may rotate through the axis (Rotate itself)
    - Flip-flopping from ER Lumen to cytosol (as well as the another direction) is very rare
12. Composition of Membrane regulates the Degree of it's fluidity
    - The more closely packed the Hydrocarbon Tails, the less fluid is the membrane
    - Shorter Hydrocarbon Tails and Unsaturation increases fluidity of membrane
    - Cholesterol in the membrane makes it less deformable (hard to move), decreasing permeability (osmosis rate) (Cholesterol cannot decrease the fluidity of membrane)
13. Cholesterol
    - Associated (Combined) with Saturated Lipids (I.E. Sphingolipids)
    - Form rigid structure and decrease fluidity
    - It can form lipid rafts (A group of lipid packed together)
    - Bent tail prevents lipid from packing too tightly
    - Act as a buffer for temperature changes in membrane fluidity

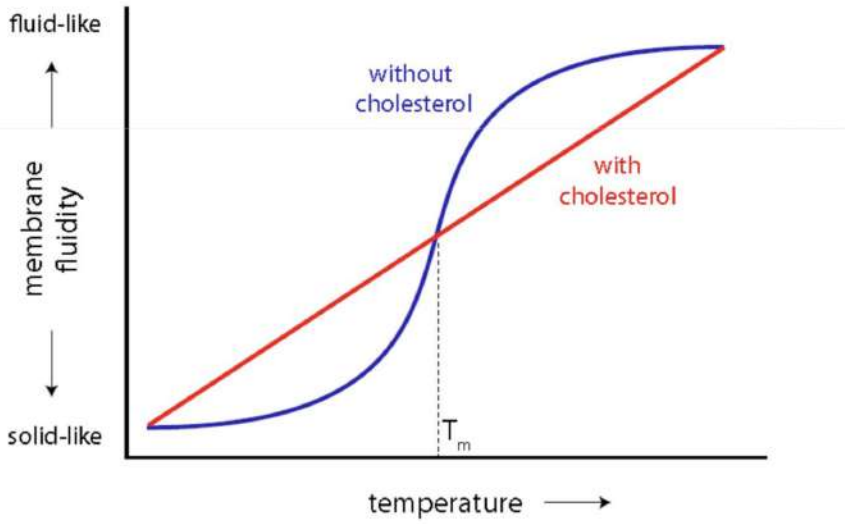

Figure 3: Temperature with respect to Membrane Fluidity

14. Explaining the Graph
    - Below T_m, cholesterol can prevent the lipid from packing tightly, which increase the fluidity of membrane
    - After T_m, cholesterol reduce the fluidity by restricting lipid movement, keeping the membrane less fluid than without cholesterol (Finally become solid)
    - Conclusion : Cholesterol cannot decrease the fluidity of membrane (Too precise)
15. Measure lateral diffusion rate on the membrane
    (a) We first use FRAP (Laser Beam) to bleaches (Deactivate) the fluorescent molecules in a specific area irreversibly, reducing their fluorescence to near zero, where other remain fluorescent
    (b) We called that specific area bleached area
    (c) Through time, the bleached area is fluorescent again. This is because other remain fluorescent has diffused to bleached area
    (d) We can measure the time and find out the lateral diffusion rate on the membrane

16. Membrane Protein
    - 50% of the membrane
    - Mainly is Glycoprotein : Sugars covalently bounded
    - No Flip-flopping (Tumble / Turning) across the bilayer
    - Can move laterally (Left and Right) within the membrane
    - Travel speed : Lipids > Protein, Because Lipid is smaller
17. Functions
    - Transporters (Slower than using a Channel Protein)
    - Anchors
    - Receptors
    - Enzymes
18. Ways to associate with the bilayer
    - Transmembrane
    - Membrane-Associated
    - Lipid-Linked
    - Protein-Attached
19. Transmembrane Protein usually span the bilayer using α-helices
    - 20-30 Non-polar residue
    - A lot of Hydrogen Bonds is formed to hide the polar group
    - Can use either Single pass (cross the lipid bilayer once with the single α helix) or Multipass transmembrane domain
    - Example of Single pass : Cell Surface Receptors
    - Example of Multipass : G protein coupled Receptors
20. Some membrane protein use β-sheets to cross the bilayer
    - Arranged form : Cylindrical conformation (β-barrel)
    - Hydrophilic amino acid residue face towards the pore (Ion-Channel)
    - Hydrophobic part face the bilayer

21. Spectrin : Links the membrane to cytoskeleton
    $\Rightarrow$ If remove the spectrin : Red Blood Cell become spherical (Normal : Biconcave disc)

22. The cytoplasmic side of the membrane is called the Cell Cortex
    - Meshwork of transmembrane proteins and filaments (spectrin)
    - Mechanical support for the membrane and cell shape

23. Intracellular transport of proteins
    - Transport through Nuclear Pores (Gated transport)
      (a) Nucleus $\Leftrightarrow$ Cytosol
    - Transport across Membranes
      (a) Cytosol $\Rightarrow$ Plastids
      (b) Cytosol $\Rightarrow$ Peroxisomes
      (c) Cytosol $\Rightarrow$ Mitochondria
      (d) Cytosol $\Rightarrow$ Endoplasmic Reticulum
    - Transport by Vesicles
      (a) Endoplasmic Reticulum $\Rightarrow$ Peroxisomes
      (b) Rest of the transport are all using vesicular transport (All can travel in $\Leftrightarrow$)

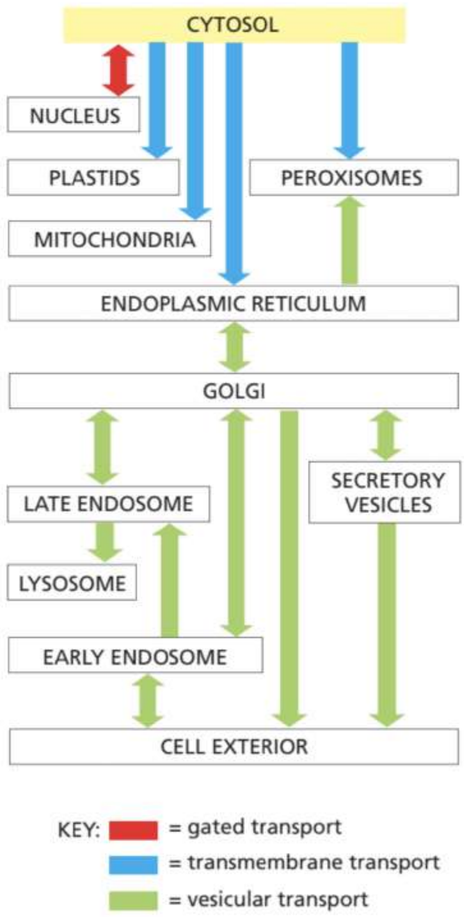

Figure 4: Intracellular Transportation of Protein

24. Sort the newly synthesized proteins by signal peptides
    Receptors are required for recognition of signal peptides

25. Ways of building protein by Sorting Signal
    - Signal Peptide (The Signal sequence will not stick together = Not Folded)
    - Protein movement after sorting signal
      Cytosol $\Rightarrow$ ER, Mitochondria, Chloroplasts, Peroxisomes and Nucleus (Those are organelle)
    - Single Peptide is functioning (Signal could be sorted) when it is linear and unfolded (3D shape does not shorting signal)

- Signal Patch (The signal sequence will stick together = Need to be folded)
  Golgi ⇒ Lysosome, Nucleus
- Single Patch required a 3D shaped (folded) before sorting its signal
- Sorting signal is similar to Translation

26. Nuclear Membrane (Nuclear Envelope)
    - Made by Two lipid bilayer with nuclear lamina and nuclear pore complexes incorporated
    - A barrier to prevent the free passage of molecules between nucleus and cytoplasm
    - Outer and Inner membrane are directly connected with the lumen of ER
    - Ribosomes are present on the outer nuclear membrane
27. Nuclear Transport
    - Nuclear Pore Complexes (NPC) : The only channel that allows transport of molecules (Ions, Proteins, RNAs) through nucleus
    - Nuclear Pore Complexes is composed of 100 $<$ distinct proteins
    - Speed : Can move 100 histone molecules = 6 ribosomal subunits per minute per pore
    - Calcium ion can move faster (Smaller)
28. Size is also a variable of being transported
    (a) Passive Transport (Diffusion)
        - Size : $<$ 20 kDa (ions, amino acids, nucleotides, small proteins)
        - $<$ 5 kDa = Fast Diffusion ; 17 kDa = 2 min ; $>$ 60 kDa = cannot enter
        - Can pass through the nucleus pore complex rapidly in both directions

- Channels Diameter : 10 nm
- Channel Length : 15 nm

(b) Active Transport

- Size : Large size (Most proteins and RNAs)
- Macromolecules contain **Special Recognition Sites**
- Energy is required (Sometimes requires transcription factor)
- Channels Diameter : At least 25 nm (Appropriate Signals)
- Unidirectional (RNA out, Protein in)

29. Nuclear Localization Signals (NLS)

- Contain a short stretch of basic amino acid residue (Arg, Lys)
- Mutation can stop the import of the protein into nucleus
- NLS/NES can be used to control the balance of a protein between cytosol and nucleus

30. Shuttling Proteins

- Shutting Protein = Nuclear Import Receptor and Nuclear Export Receptor
- Import and Export the target protein (Cargos)
- Every receptor carry 2 binding sites for cargo and FG repeats on nucleoporins
- FG is a recognised amino acid sequence that create a binding track for the transporting receptors
- Cargos can travel the nucleus and the cytoplasm by GTPase (Ran)

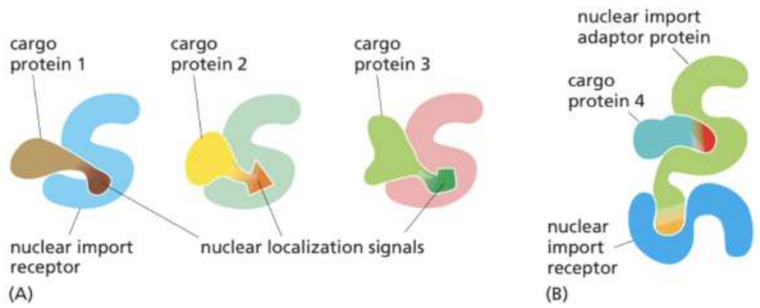

Figure 5: Shuttling Proteins Binding

- Ran = GTP Binding or GDP Binding
  (a) (Outside the Nucleus) Ran starts in the cytosol bound to GDP (Ran-GDP) in inactive form
  (b) If Cytosol contains GTP, GTPase activating protein (Ran-GAP) promotes hydrolysis of GTP to GDP
  (c) This can ensure that GDP and Ran-GDP is remain bounded in the cytosol and this will release a phosphate (p_i)
  (d) Ran-GDP will then pass through the nuclear pore complex (NPC)
  Until now, Ran-GDP is still remain inactive
  (e) (Inside the Nucleus) Ran-GEF (Guanine exchange factor) will then catalyze Ran-GDP and activate the GDP and promotes the exchange of GDP with free GTP
  (f) It will release GDP and Ran will bind with GTP (Nucleus has high concentration of GTP; we get GTP in here)
  (g) During step (f), Ran-GDP will release the cargo, which contain Importins (Nuclear Localization Signals, NLS) to nucleus at the same time
  (h) Also, it will also bring something (Exportins) away the nucleus by binding to cargo (mRNA, proteins with Nuclear Export Signals, NES)
  (i) Ran-GTP stabilizes this complex and pass away through the nuclear pore complex (NPC)

(j) Once the export complex reaches the cytosol, Ran-GAP again promotes the hydrolysis of GTP to GDP on Ran, converting Ran-GTP back to Ran-GDP

- All Protein with the nuclear localization signals need to be regulated by phosphorylation and dephosphorylation or dissociation of inhibitory subunit before they are imported, even though they are already synthesized
- The process of the whole cycle is Active Transport
- The movement of the Ran protein itself is Passive Diffusion

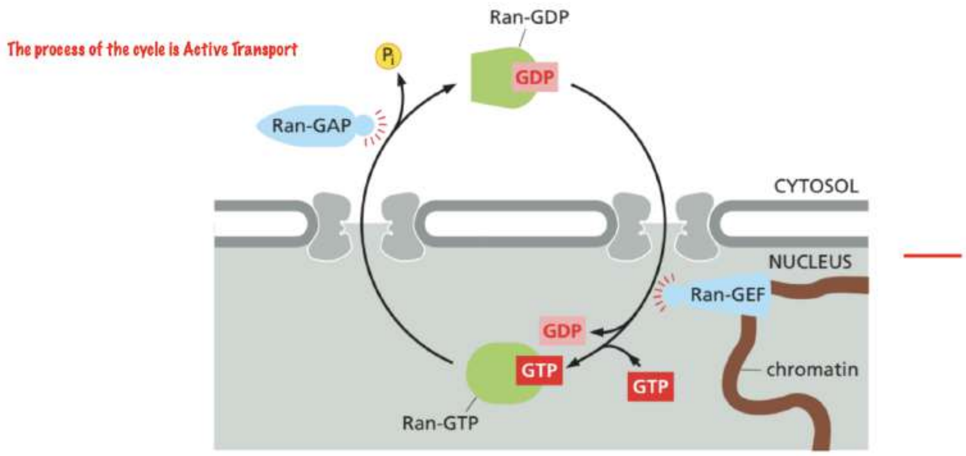

Figure 6: Ran Cycle

31. Transport of RNA out of the nucleus
    - Export RNAs (tRNAs, mRNAs, rRNAs) requires energy and GTPase (Ran-GAP)
    - RNA is exported as a large RNA-protein complex, but not naked RNAs (It is naked when it is remain in nucleus)
    - Pre-mRNAs and mRNAs are associated with > 20 proteins to form hnRNPs before it is released
    - At least one protein contain NES (nuclear export signal)
32. Ribosome = Protein Import + Ribosome Export
33. Nucleoi = Produce ribosome subunits

34. Nucleoi = Multi-phase membraneless organelles cut through liquid-liquid phase seperation
35. Liquid-liquid phase seperation : Membrane Partitioning (Membrane separation)
36. Nuclear Lamina
    - 4 different lamins : A, B1, B2, C
    - Molecular weight : 60-80 kDa
    - Homology to the intermediate filament proteins of cytoskeleton
    - B-type lamins is covalently linked with a lipid molecule (Prenylation) at its C-terminus cysteine residues
    - It attaches to the inner nuclear membrane through prenylation and interaction with integral membrane proteins
    - The nuclear lamina serve as a site of nuclear attachment
37. Breakdown and Reformation of nuclear envelope : Mitosis
38. Phosphorylation of lamins ⇒ Disassembly of the nuclear lamina
39. Dephosphorylation of lamins ⇒ Fusion of nuclear envelope vesicles

2 Lecture 3 & 4 Techniques in Plant Cell Culture I

1. Organelles in Plant cells
   - Chloroplast : Photosynthesis (Absent in onion epidermis)
   - Mitochondria : Respiration + Obtaining energy from sugar metabolism for ATP
   - Nucleus : Control centre + Nucleotides + DNA
   - Cell Wall : Structural support
   - Large Vacuole : Biggest Organelle + Provide turgor pressure
   - Endoplasmic reticulum (ER) : No streaming
   - Spherosomes/ oleosomes : Single membrane + 1mm Oil Droplet
2. Function of Vacuole
   - Bound by vacuolar membrane (Tonoplast)
   - Large Fluid-Filed Vesicles
   - Allow cells to adjust size and turgor pressure
   - Contains inorganic ions, sugars, enzymes
   - Storage + defense
3. Function of Mitochondria and Chloroplasts
   - Made by ancient symbiosis
   - Before : Aerobic Prokaryotic and establish to an endosymbiotic relationship mitochondria
   - Eukaryotic cells contain mitochondria and engulfed photosynthetic prokaryotes (Chloroplast)
4. Function of Cell Wall
   - In Cellulose
   - Rigidity and Strength
   - Can against the turgor pressure
   - Provide a barrier for individual cell

- Formed by β 1-4 linkage Glucose molecules
- Fully Impermeable

5. Function of Plasma Membrane
   - Permeable to water and gases
   - Impermeable to ions and large molecules
   - Maintain a different internal environment from surroundings
   - Water moves across a semipermeable membrane from lower osmotic concentration (Higher water potential) to high osmotic concentration (Lower water potential)
6. Turgor Pressure
   (a) A potential internal hydrostatic pressure within a plant, fungal or bacterial cell
   (b) Pushes the plasma membrane against the rigid cell wall
   (c) Plasmolysis : Loss of water $\Rightarrow$ Loss in turgor pressure
   (d) Osmosis : Re-establish the turgor pressure (Water moving into plant cells
   (e) Flaccid cells : Water loss + No turgor pressure
7. Plasmodesmata : Interconnect cells through cell wall openings (Pits) + Connect to the plasma membrane

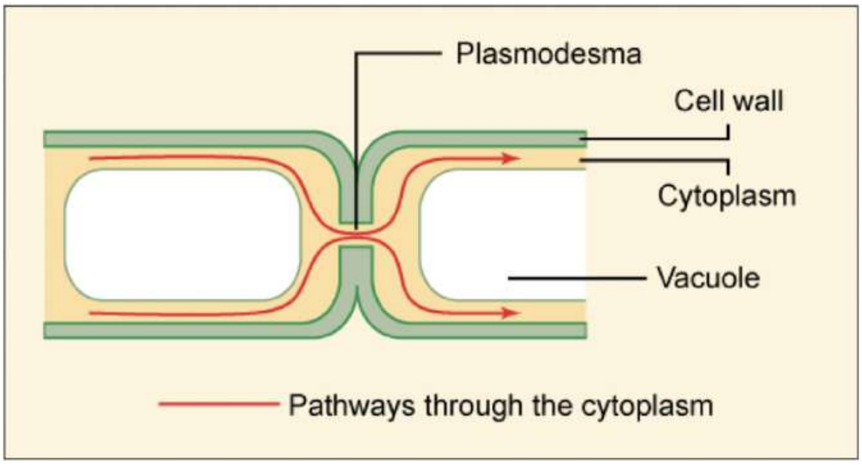

Figure 7: Plasmodesmata Function

8. Protoplast : Plant cells lack of cell wall
   - Heterogeneous in size and content
   - Enzymes (Cellulase & Macerozyme) : Digest cell wall
   - Shape : Usually spherical (Because of non-directional osmotic forces)
   - Osmotic Force = The only thing to maintain the spherical shape

Figure 8: Osmotic Force

   - Remaining structures : Bounded by the plasma membrane
   - Some may not be spherical shape ($\because$ Have internal cytoplasmic strands)
     $\Rightarrow$ Protoplast only remove the external support

9. Cytoplasmic Strands
   - Filament-like extensions of cytoplasm
   - Composed of actin-filaments and myosins (cytoskeleton)
     $\Rightarrow$ Reorganizing the strands constantly for cytoplasmic streaming and maintain cell health
   - Function 1 : Connecting plant cells through plasmodesmata
     $\Rightarrow$ Facilitate communication and transport (Ions, Nutrients)
     $\Rightarrow$ Holding the nucleus and other organelles within the cell
     $\Rightarrow$ Acting as dynamic pathways for organelles and molecules within the cytoplasm
   - Function 2 : Restrict the distribution of organelle
   - Function 3 : Organize the vacuole and cytoplasm (Organize the size)

- Function 4 : Enabling cytoplsmic streaming
- Function 5 : Control the shape when it is non-spherical shape

10. Cytoplasmic streaming
    - Directional movement of cytoplasm converging at nucleus
    - Optimize the transportation and the distribution organelle (Act as a cargo)
    - Enhance enzyme reaction
    - The strand length, branching and attachment change dynamically
    - If the strand is cut, network collapses without destroying cell integrity (nothing change for the cell morphology)
    - Osmotic forces can take over and act round the protoplast
    - Cytoplasmic streaming still functions in intracellular transport (Not strong enough to hold the extracellular shape)
11. Transvacuolar strands (TVS)
    (a) Shape : Thin & Tube-like
    (b) Formed by the vacuolar membrane and it can cross over the large central vacuole in plant cells
    (c) Function 1 : Act as cytoplasmic bridge in **cytoplasmic streaming**
    (d) Function 2 : Increase the surface area between cytoplasm and vacuole for material exchanging
    (e) Function 3 : Connect the peripheral (surrounding) cytoplasm and the nucleus
    (f) Function 4 : Organize the vacuole and cytoplasm (Organize the size)
    (g) Function 5 : Connecting distant cytoplasmic regions
    $\Rightarrow$ Can enlarge the molecular, organelle, or signal interactions which occurs within cells or tissues

12. Specialized Plant Cells : Undergone differentiation to develop a unique shape, structure, and composition
    ⇒ Allowing them to perform a specific function with high efficiency
13. Trichomes : From leaves of model plant Arabidopsis
    (a) Fuzzy Leaf : Trichomes / Hairs
    (b) Function : Protect against Pests & Reduce Evaporation
    (c) Two Types of Trichomes : Non-Glandular Trichomes & Small Glandular Trichomes
14. Guard Cells : A pair of specialized cells that form Stomata for gas exchange (as well as allow pathogen entering)
15. Stomata : The door of cells for gas exchange
16. Xylem : Specialized water-conducting cells
    - Form thickened secondary wall rings
    - Dead cells and tissues
    - Hollow and reinforced vessel

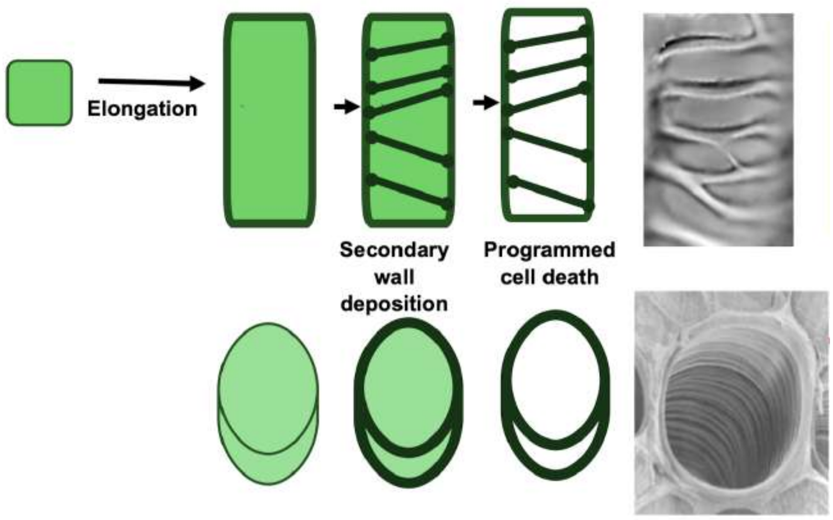

Figure 9: Xylem Structures

17. Phloem : Companion cell (CC) and sieve element (SE)

(a) Companion cell (CC)
   - Link the sieve element
   - Provide genetic materials through extensive plasmodesmatal connections

(b) Sieve elements (SE)
   - The functional one lack nuclei and central vacuoles
   - Facilitating flow
   - End of adjoining sieve element : Connected by pores
   - Pores : Formed by enlarged plasmodesmata
   - ATP can also transported through plasmodesmata (Hole)

(c) Sink tissue : A new tissue that can store carbohydrates

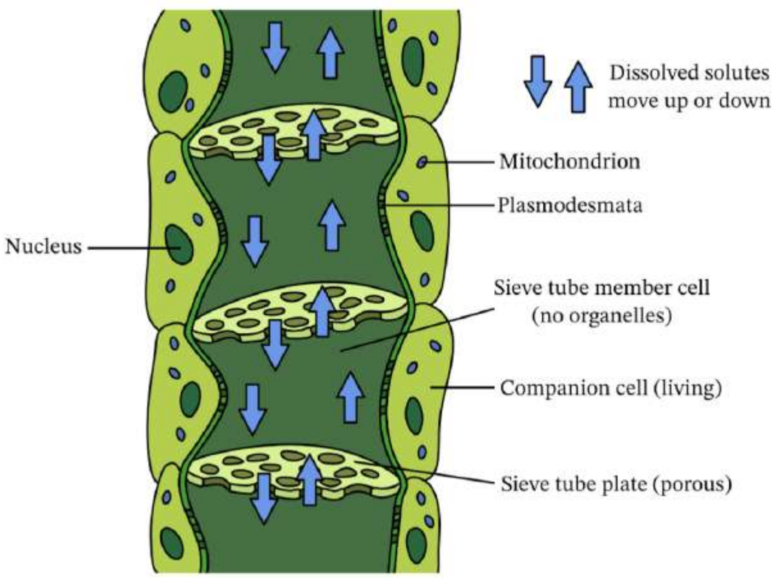

Figure 10: Phloem Structure

Techniques in plant cell biology

1. Robert Hooke discover cells under light microscope
2. Type of microscopes
   - Light Microscopy
   - Electron Microscopy (Scanning / Transmission)

- Confocal Laser Scanning Microscopy (CLSM)
  $\Rightarrow$ Require fluorescence ; Microscope can active it and show image

3. Light Microscopy
   (a) Fixed specimen in embedding mould
   (b) Cut sample by a rotary microtome (No distortion in this method)
   (c) Block the sample (stable storage)
   (d) Staining or immuno-histochemical labeling method is required
4. Properties
   - Cannot show the process of a plant (Like growth and production
   - Only **Limited growth structure** could be observed
   - Have time lapse of growth
   - Need to kill the plant
5. Immuno-histochemical staining
   (a) First, add Anti-mouse Antibody and conjugated with alkaline phosphatase or Horse Radish Peroxidase
   $\Rightarrow$ Can emit signal and detected by microscopy (E.g. Fluorescent (Colour Signal))
   (b) Then, bind an Anti-mouse antibody (Goat)
   $\Rightarrow$ It is a protein and cannot observe by our eye directly
   (c) At last, bind a Specific Antibody (mouse)
   (d) Coloured or fluorescent signal released from reacting with substrate
6. Properties
   - Sectioning & Immuno-staining
   - Cell sacrificed in immunostains (This method will kill the cell and cannot function anymore)

- Labour (Manual) and need time

7. Electron Microscopy
   (a) Electrons replace light (Can increase resolution)
   (b) Can see a smaller object (Subcellular Organelles)
   (c) Also can immunostained (Use gold particles and bind to the antibody)
8. Properties
   - Require more time than Light Microscopy
   - cell sacrificed
   - No dynamic motion
9. Confocal Laser Scanning Microscopy
   (a) Use laser on specimen to produce optical slices of tissue (from top to bottom layer by layer
   (b) Can us live cells that are auto-fluorescent
   (c) Confocal image showing localization of the auto-fluorescence GFP-tagged fusion protein instantly
10. Localization : Maintain transport things in a specific location
11. Properties
    - No actual slicing
    - No sample sacrificed
    - Provide 3D images
    - Can provide a time series of event (We can know how the plant grow under microscope)
    - Can use different fluorescent-tagged probes at the same time (Can identify the organelle)
    - Have time lapse of dividing cells
    - Need computer to observe
12. Comparison of Confocal and Light Microscopies

- Confocal microscopy & GFP tagging shows movement of vesicles through a root hair
  ⇒ Fastest Growing Plant Cells : Root Hair
  ⇒ Vesicles delivered to plasma membrane by motor proteins along cytoskeleton
- Confocal requires new membrane and cell wall materials
- Confocal need to synthesized in ER, Modified in Golgi apparatus and Packed in vesicles

3 Lecture 5 & 6 Techniques in Plant Cell Culture II

1. Plant Tissue Culture
   - = In vitro plant culture (Outside the plant)
   - We do it in an aseptically culture and liquid or on solid (Agar) medium with suitable nutrients
   - Fate 1 : Propagates as a mass of undifferentiated cells for a period of time
   - Fate 2 : Regenerated into a whole plant
2. Workflow of plant tissue culture :
   Explant (A cell or a tissue) $\xrightarrow{\text{Initiation}}$ Callus (Healing tissue) $\xrightarrow{\text{Proliferation}}$ Shoot formation & Root formation $\xrightarrow{\text{Pre-transplant}}$ Plantlet (Young and Small Plant) $\xrightarrow{\text{Establishment}}$ Plant
   (a) Initiation
      - Cut, Sterilize media
      - Start growth
   (b) Proliferation
      - Growth : Explant $\rightarrow$ Plantlets
      - Cut and multiplication
      - Explant undergo dedifferentiation to form a callus
   (c) Pre-transplant
      - Plantlet growth + Initiate root & shoot development
      - Transfer into favourable media and condition for growth
   (d) Establishment
      - Transplant of conventional growth condition, e.g. Soil
3. Function of Plant Tissue Culture
   (a) Essential technique in plant biology
   (b) Commercial production of disease-free valuable plant by micropropagation

⇒ E.g. Multiplying stock plant(Mother) rapidly to produce progeny plants (Offspring)

4. Advantage of plant tissue culture over intact plants (perfect plants)
   (a) Grow plant cells in liquid culture (Bioreactor) for the commercial production
       - Not affecting the productivity
       - No endangering wild plant
   (b) Shorten time
       - Obtain homozygous in breeding by the production of di-haploid plant from haploid cultures
       - Use for food and bio fuel
   (c) Regenerate novel hybrid species (overcome reproductive isolation
   (d) Feasibility in screening large numbers of plant lines in limited space
       - Achieve salt/herbicide-resistant (Remove grass) variants
       - Screening program of cell (not plants) for advantageous characters
   (e) Uniform production from limited material
   (f) Exclusion of pests (fungi, bacteria, viruses, insects, nematodes)

Techniques for plant tissue culture and cell technology

1. Sterilization and aseptic techniques : Prepare for a microbial-free (aseptic) environment
   (a) Aseptic tool
       i. Prevent unwanted microorganisms like bacteria, fungi, and viruses from contaminating the growth medium and explants
       ii. Microorganism : Compete for nutrients, inhibit plant growth, reduce success rates, damage the plant tissues, and ultimately compromise the entire culture

iii. Tools
   A. Culture vessels (Petri dish, Plastic containers, Flasks)
   B. Autoclave to sterilize containers and media (Kill by high pressure)
   C. Scalpels, blades, forceps sterilize by flaming or by using dry-heated glass beads in a heat sterilizer
   D. Laminar flow cabinet (fitted with UV lamps) for aseptic transfer
   E. Shakers for suspension cultures
   F. Shelves for containers

iv. Sterilize by Heat
   A. Wet Heat : $120°c + 103.5kPa + 15 - 20$ minutes $\rightarrow$ High Temperature + High Pressure + Short Time
   B. Dry Heat :
      - Oven : $160°c + 1hour \rightarrow$ High Temperature + Long Time
      - Frame : $600°c \rightarrow$ Very High Temperature

v. Sterilize by Chemical : 70% Ethanol + 1% Na-hypochlorite

vi. Sterilize by Filtration : Filtrate away the large contaminator (Pore size)

vii. Sterilize by Irradiation : UV light

(b) Aseptic technique
   - Operator's hands must be washed
   - Use High Efficiency Particulate Air (HEPA) : A Filtration Machine
     – Laminar flow hood has HEPA-filtered air
     – Working surface must be **swiped with 70% ethanol**
     – Structure : Filter sheet of randomly arranged fibres
     – High density of the filter frame $\rightarrow$ can capture 99.7% of airborne particles by the HEPA-filter
   - Steps :
     i. Room air enter when you place the sample onto the working stage

   ii. Contaminated air will then enter stage 2, where can clean the contaminated air
   iii. Clean air will go to stage 3 after the filtration
   $\Rightarrow$ The HEPA Filter can confirm all air is clean
   iv. At the same time, clean air is releasing to the environment
   v. At last, Clean filtered air will enter stage 4, where is our working stage
   - Before we use the HEPA filter, we need to test by white smoke (Crucial Part)
   $\Rightarrow$ Test how the air goes when the cabinet is first used (Evaluate the air flow)
   $\Rightarrow$ Make sure the air is cleaned after enter HEPA filter $\rightarrow$ Avoid environmental contamination

2. Preparation of Media

   (a) Medium

   i. Use Murashige and Skoog Medium (MS Medium) $\rightarrow$ Heat-labile (Not resist to heat)
   ii. Solution : Add the medium after autoclaving by filter-sterilization (Recall : Pressure)
   iii. Optimal pH : 6.0 (Acceptable Range : 5.0-6.5); Must $<$ 7.0
   iv. Reason : It will affect the nutrient intake in the roots and the plasma membrane (E.g. Nitrogen (Macronutrients), Iron(Micronutrients))

   (b) Media Composition

   - Macronutrients
     - Major are inorganic nutrient (Amount : 10^{-3})
     - Function : Support Nucleic acid and Protein synthesis
     - Nitrogen : Obtain from nitrate and Ammonium
     - Phosphorus : Obtain from Phosphate
     - Sulphur : Obtain from Sulphate

  - Potassium : Function : cell growth
  - Calcium : Function : Cell wall synthesis/ formation
  - Magnesium : Function : Enzyme co-factor + Membrane integrity
- Micronutrients
  - Trace element (Small amount) (Amount : 10^{-6})
  - Iron (Fe) (Enzyme co-factor)
  - Manganese (Mn)
  - Zinc (Zn) (Enzyme co-factor)
  - Boron (B)
  - Copper (Cu) (Enzyme co-factor)
  - Cobalt (Co)
  - Molybdenum (Mo) (Enzyme co-factor)
- Vitamins
- Sucrose (Get carbon)
- Organic Nitrogen
- Agar (Solidifying agent)
- Extra mixture (Optional)
  $\Rightarrow$ E.g. Coconut milk, Yeast extract $\rightarrow$ Promote cell division and somatic embryogenesis
- Plant Growth Regulator (Optional)

(c) Plant Growth Regulator

- Hormone synthesis : Affect growth & development (Synthesis derivative still works)
- If Explant Produce enough Auxin and cytokinins by itself, then additional plant growth regulator
- Stock solutions (Indole-3-acetic acid = IAA): Store in the dark (Some auxin is light-intensive)
- Plant cell : Totipotent (Stem Cell) $\rightarrow$ Can regenerate and differentiate into any cell type

(d) Concentration ratio between Auxin & Cytokinin

i. Auxin = Cytokinins = Low $\Rightarrow$ Minimal cell proliferation

ii. Auxin = Cytokinins = High ⇒ **Callus**
iii. Auxin = High; Cytokinins = Low ⇒ Send out roots (Lower Part)
iv. Auxin = Low; Cytokinins = High ⇒ Send out shoots (Upper Part)

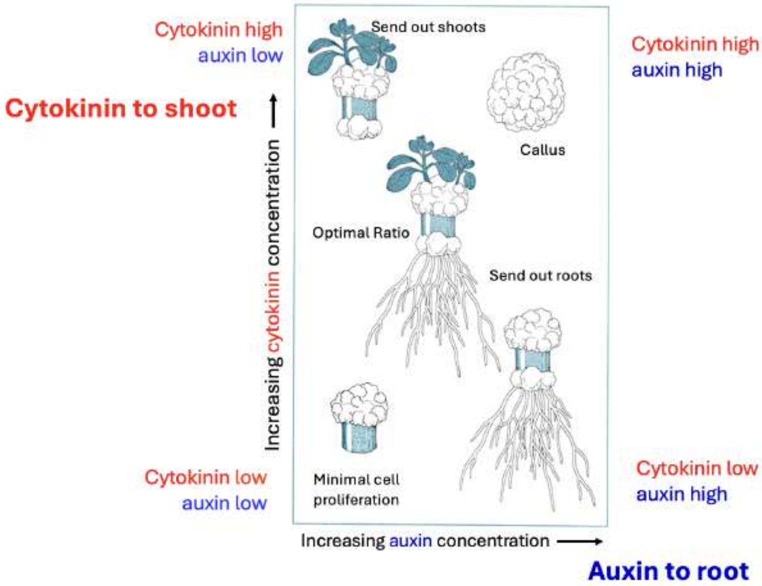

Figure 11: Concentration Ratio of Plant Growth Regulator

(e) Example of Auxin

- IAA (Natural occurs) : For callus induction
- IBA : Rooting shoots
- NAA : Synthetic analogous of IAA
  ⇒ Lower Concentration : Roots Induction ⇒ Higher Concentration : Callus Induction
- 2,4-D : Synthetic auxin for Callus Induction & maintenance

(f) Example of Cytokinin

- BAP : Lower Concentration : Callus induction ; Higher Concentration : Shoots Induction

3. Selection of Explant

(a) In theory, we can select any part of the plant (Because all are totipotent)
(b) Except Xylem and Phloem ($\because$ Dead cell + Lack of nucleus)
(c) Reality, we select the young (more sensitive respond) + disease free + actively divided tissue
(d) We need healthy growing living cells

4. Surface Sterilization
   (a) Removing microbial contamination $\rightarrow$ Cannot use high temperature (Will kill the explant)
   (b) Surface Sterilization : 70% ethanol or 1% bleach + 1% sodium hypochlorite
   $\Rightarrow$ Must wash in sterile water (Because it is toxic to plant)

5. Establishing in Vitro Culture
   - Types of in Vitro Plant Culture
   (a) Intact (Perfect) Plant Culture
      – Sowing of orchid seed in vitro $\rightarrow$ Seedling and then whole plant
   (b) Callus Culture
      – Occurs on cut surfaces $\rightarrow$ gradually cover $\rightarrow$ Heal the Damaged area
      – Dedifferentiate the **Colourless mass of cells** (Explant + mass of cell = A group of cell) from differentiated tissue to form callus cells
      – Solid medium (Agar) : Callus cells can stick together
      – Liquid medium : Can occur as individual cells or in small clusters $\rightarrow$ Form Suspension Culture

Figure 12: Colourless Callus cell

- Optimal Case : Plant can be maintained indefinitely if necessary nutrients are given without contamination
- Must be sub-cultured (Reproduce) regularly into fresh growth medium
  ⇒ Growth medium affects the re-differentiation (Some can re-differentiate into a whole plants
- Technique in callus tissue culture
  i. Initiation of culture : Explant is aseptically incubated on medium
     ⇒ Reason: Medium can provide necessary nutrient to explant

ii. Concentration of the growth regulators (auxin and cytokinin) is significant in initiation and development of culture (Mentioned before)

iii. Choose a explant species that is easy to regenerate $\Rightarrow$ E.g : Tobacco

(c) Embryo Culture : Isolate embryos, used for callus induction

(d) Organ Culture : Isolate organ, used for explant culture $\Rightarrow$ E.g. : Meristem, root, ovule

(e) Suspension Culture : Keep changing new medium

- Derived from a tissue or callus
- Cannot stop
- Require a shaker
- Incubate at room temperature
- Can grow faster than callus culture

(f) Protoplast Culture

- (Recall) Protoplasts (Totipotent): Plant cells removed of their cell walls but maintain their cytoplasm
- Protoplast culture : Isolating protoplast from organ by enzymatic digestion of cell wall, maintaining them in a nutrient Liquid medium
- Goal : Regenerate cell walls + Induce cell division
- Reason :

i. Ability to regenerates to one plant

ii. Useful for somatic hybridization (Protoplast fusion)

- Challenging :

i. No guarantee that can sustain division and regenerate from protoplast

ii. Protoplast embryos has better regeneration potential than leaf

(g) Scaling–up

- Use of bioreactors/machine : Air-lift fermenter, Stir tank fermenter

  - Lab scale : 5L; Production Scale : 10-30L
  - Limitations of bioengineering problems
    * Plant cell are large
    * Cells like to stick together
    * Cells can be damage easily by cutting it
    * Only grow at high density
    * Some have slow growth rate
- Technique in Vitro Plant Cell Culture
  - Aseptic Technique
  - Initiation of Callus Cultures
  - Regeneration of Tissue Cultures

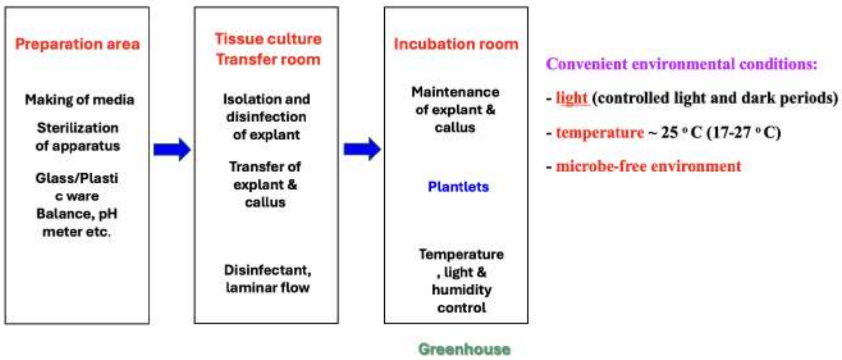

Figure 13: Environmental Condition

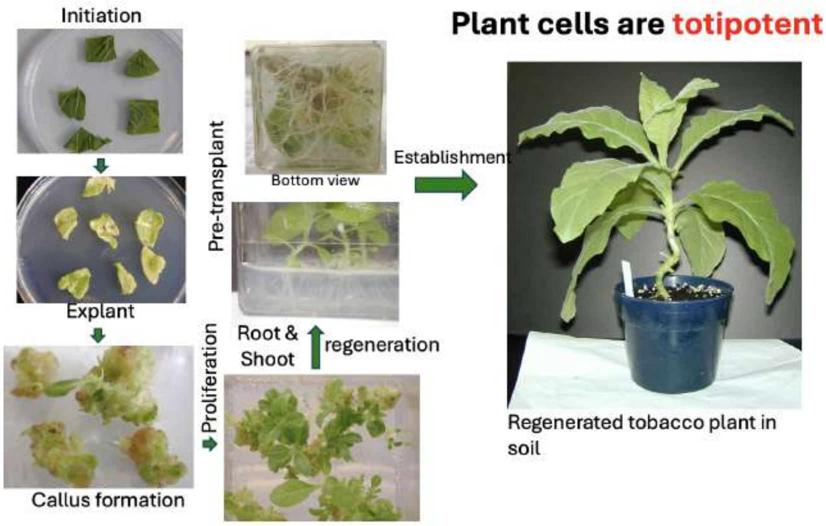

Figure 14: Example of the whole process

4 Lecture 7 & 8 Techniques in Plant Cell Culture III

1. Micropropogation : Using extremely small pieces of plant tissue and growing these under laboratory conditions to produce new plants
   ⇒ Used in commercial horticulture
   ⇒ Require longer time if we don't use micropropogation
   ⇒ Steps is introduced above

2. Haploid Culture
   (a) A in vitro culture method where we use haploid cells (Pollens or Ovules)
   (b) Not diploid (Because Parents are haploid)
   (c) Monohaploid has only 1 set of chromosomes → Sterile (No Baby)
   (d) Production of diphaploid plant from haploid plant : Can get homozygous lines
   (e) Anther culture is one of the haploid plant production method
   ⇒ We choose the open flower bud → Pollen already spread our
   ⇒ Reason : Because One Anther has many spores (Many starting material)
   (f) Colchicine treatment : Achieve homozygous dihaploid → A mitotic inhibitor that can prevent microtubule formation and inducing polyploidy
   ⇒ It cut the mitosis process and form the nuclear envelope again after the replication phase.
   ⇒ We can get two haploid chromosomes and form a homozygous dihaploid (Because they have exactly the same genetic information)

3. Protoplast culture : Protoplast are isolated plant cells without cell wall
   (a) PEG Mediated Transformation : A simple and efficient method to transform moss (Used Microinjection)

- Simple and efficient method to transform moss

(b) Somatic Hybridization (Protoplast Fusion) :
- Can only work when the cell wall is gone
- Any combination of protoplast can be induced undergo fusion
- Allow generation of hybrid between sexually incompatible species
- Can genetic modified sterile species
- Can form heterokaryons cells
  $\Rightarrow$ A single cell contain two or more genetically different nucleus
- Two nuclei inside the heterokaryon eventually fused $\rightarrow$ a hybrid nucleus
  $\Rightarrow$ Then stimulated to divide, form a new cell wall, and develop into a callus

(c) Step of fusion
  i. Isolation of protoplasts : $A + B$
  ii. Fusion of protoplasts (Need Polyethylene glycol) : $A + B$ (AB)
  iii. Regeneration from selective tissue : AB
  iv. Analysis of regenerated plants : AB

4. Somaclonal Variation to make new varieties
   $\Rightarrow \because$ Asexual reproduction has a Mutation Risk
   - Somaclonal Variation : Might cause Genetic Variation due to genetic instability
     $\Rightarrow$ E.g. Change in chromosome number & structure, DNA sequence
   - Pros : Increase the feasibility of novel variation (Favorable mutation)
   - Cons : $\because$ Asexual reproduction (Might cause Genetic unstable)
     $\therefore$ Chance of Somoclonal Variation

- Factor affect mutation
  (a) Method of Propagation
    - Shoots induced by Regulator (Increase mutation)
    - Single-node / Axillary-bud method for propagation (Reduce mutation)
  (b) More subculture, Increase mutation (Higher chance to get with)
  (c) Choice of Starting Material : Use undifferentiated tissue (Reduce mutation)
  (d) Auxin, Cytokinin Regulator used : NAA (Increase mutation)

5. Grafting different plants to make new varieties (Fusion of adjacent tissue together) : Stem-to-Stem Grafting
   - Have genetic communication
   - Create new polyploid species by transferring whole nuclear genomes
   - Crops : Allopolyploid = Contain > 2 chromosome sets from different species by sexual hybridization
   - After fusion : Graft site is cut and cultured (Plant resistant to both antibodies)
   - Number of Chromosome : Addition of two plants
   - Morphology and phenotype changes (Has intermediate traits)

6. Preservation of germplasm
   - Plant embryos can be stored
   - Germplasm can be stored and micropropagated when needed

7. Cryopreservation = Low temperature storing
   - Use -196°c liquid nitrogen to store
   - Can store for long period
   - No cellular activity during preservation
     $\Rightarrow$ Might lead to cell death if the storing time too long

- Need to thaw : Maintain the cell life (Set a time)

8. Production of secondary metabolites from plant culture
   (a) Primary Metabolites : Sugar, Amino Acid, Nucleotides, Chlorophyll
   (b) Second Metabolites : Natural Products (Useless in growth & development
   $\Rightarrow$ Used to Protect against Predation & Pathogen
   $\Rightarrow$ E.g. Dyes, Flavor Compounds
   (c) Can get from the plant cells which grown in bioreactor

9. Function of Tissue Culture Technology
   - Use clonal selection choose metabolite
   - Coloured metabolites can be selected visually (Easy to be distinguished)
   - The clone must be checked for stability
   - Conditions for cell culture must be optimized (Minimize the error)

10. Production of metabolites from root culture
    - Roots of plants naturally produce secondary metabolites
    $\Rightarrow$ Reason : Interaction against with pathogens and predators
    - ***Agrobacterium rhizogenes*** : Cause Hairy Root Disease = Induce Root Formation
    - Reason : Contain root-inducing (Ri) plasmid
    - Can use for natural product synthesis (Incorporated into plant genome)

11. Production of Taxol
    - Anticancer drug (Inhibiting cell division and prevents cell migration)
    - Abundant in the needles of the Pacific yew (Renewable resource)

- Tissue culture initiated by screening bark tissues of Pacific yew
  $\Rightarrow$ Identify best producers
- Cannot use in bacteria ($\because$ Most bacteria often lack complete metabolic pathway)

12. Large-scale plant cell culture is limited
    - High cost
    - Long time investment
    - Medical plant has unique metabolic flow and metabolite (Hard to find the genomic information in short time)
    - Cheap production essential (Use plant cell as bioreactor)
      - Cell feed a particular complex metabolite $\rightarrow$ convert into high-valued product (Need one or two crucial enzymes) $\rightarrow$ Enzyme can act on many plants
13. ***Agrobacterium tumefaciens***
    - Plant pathogen forming tumors on dicots
    - Can enter plant through wound sites (Hurting part)
    - Contain Ti plasmid (Tumour-inducing) $\rightarrow$ Cause Crown Gall disease
    - Ti plasmid : Contain T-DNA $\rightarrow$ Integrated into plant genome
    - Method to introduce recombinant gene into plant:
      - Microinjection
      - Polyethylene glycol (PEG)
      - Particle bombardment (Use high pressure + DNA Particle Gum)
        $\Rightarrow$ Contain higher resistant to something that you want to inhibited
14. Biopharming (Making medicine biologically) : Genetically engineered crops
    $\Rightarrow$ Produce pharmaceuticals = Drugs = Medicines

15. Advantages

- Reduce production cost
- Can use food as vaccine delivery system
- can use food as drugs
  ⇒ E.g. Banana, Carrot

5 Lecture 9 & 10 Transport across the Plasma Membrane

1. Plasma Membrane : Barrier
   - Bilayer : Barrier to the passage of various polar molecules
   - Generating ionic concentration differences across the lipid bilayer
     $\Rightarrow$ Different concentration within the cell
   - Potential energy is stored in the form of electrochemical gradients
     (a) Generate high energy molecules from membrane-enclosed organelles
     (b) Drive transport processes
     (c) Convey electrical signals in electrically excitable cells
2. Passive Diffusion : Molecules that can diffuse through membrane
   - Hydrophobic molecules
   - Small Uncharged Polar Molecules
   - Large Uncharged Polar Molecules
   - More charged or polar, The harder to cross the membrane
3. Molecules cannot diffuse through membrane : Ion
4. Importance of crossing membrane: Polarity, Charged $>$ Size
5. Permeabilities of various biomolecules
   - Permeable : Depends on polarity, size
   - Non polar (Lipid soluble) molecules : Can cross the lipid bilayer rapidly
     $\Rightarrow$ E.g. Steroid hormones, oxygen, cholesterol
   - Polar, Uncharged molecules : Small molecules can pass (Large molecules cannot)
     $\Rightarrow$ Small : Water, ethanol
     $\Rightarrow$ Large : Glucose

- Charged molecules : Cannot pass through the lipid bilayer

6. Passive Transport vs Active Transport
   - Passive : Move from high concentration area to a low concentration area without consumption of extra-energy (2^{nd} law of thermodynamics)
   - Active : Move from low concentration area to a high concentration area
     $\Rightarrow$ Additional energy source is needed
7. Driving force in passive transport
   (a) Uncharged molecules : Only Concentration Gradient
   (b) Net charged molecules : Both Concentration Gradient & Membrane Potential
   (c) Concentration Gradient + Membrane Potential = Electrochemical Gradient
8. Transport Protein
   - Transporter
     - Can move
     - Used for transport amino acids or sugar
     - Working Principle : Contain a Solute-binding site on one side, changing the shape (conformation change) and release it on the other side
     - Can be either Passive or Active
   - Passive Membrane Transporter
     - Channel Mediated
     - Transporter Mediated : Energetic Favorable
   - Active Membrane Transport
     - Coupled Transporter : Exchanging ions
     - ATP-Driven Pump : Control the phosphorylation by energy ATP
     - Light-Driven Pump : Emitting light

- Channel Proteins
  - Only passive transport
  - Function : Allows the passive movement of inorganic molecules
  - Have a hydrophilic pore for specific ion
    $\Rightarrow$ Allow specific ions to pass through ($Na+$,K^+,Ca^{2+},Cl^-)

9. Transport speed : Channel Protein $>$ Transporter Protein

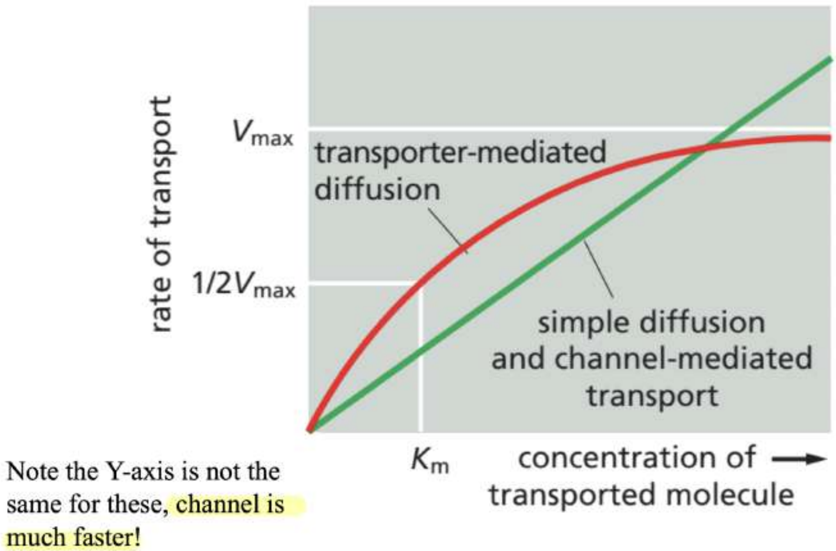

Figure 15: Kinetic of Simple Diffusion vs Carrier-mediated Diffusion

10. Coupled Transport : Energetic Favorable

- Uniport : Only one transported molecule is needed (Not a coupled transport)
- Symport : Binding one transported molecule and one co-transported ion
  $\Rightarrow$ E.g. Na^+/Glucose symporter : Both Na^+ + glucose are entering to the cytosol against its concentration gradient (Use ATP but not directly)
- Antiport : Bind two molecules in opposite direction

- Symport and antiport are coupled transport

11. Secondary Active Transport : Use energy ATP indirectly
$\Rightarrow$ E.g. Na^+ K^- Pump = ATPase operating as an antiporter

12. Na^+ K^+ Pump

    (a) 3 Na^+ are pumping out the membrane

    (b) 2 K^+ are pumping in the membrane

    (c) Inside : Higher concentration of K^+

    (d) Outside : Higher concentration of Na^+

    (e) Function : Create **electrochemical gradient, maintain osmotic balance and cell volume regulation of cells**
    $\Rightarrow$ Osmotic pressure will force the water in and out of the cell

    (f) Net loss of one positive charge from the inside of the cell every pump
    $\Rightarrow$ Not the major thing that make the membrane potential

    (g) Spend $\frac{1}{3}$ of the energy in cell on this pump

13. Ca^{2+} ion Pumps

    (a) Maintain low intracellular level of Ca^{2+}ions

    (b) Use Ca^{2+} as a signal need to be very low concentration
    $\Rightarrow \because Ca^{2+}$ ion is toxic

    (c) Directly use 1 ATP to pump out 1 $Ca^{2+}$of the cytosol, exchanging 3 Na^+ coming in

    (d) This exhange also depending on the gradient of Na^+K^+ Pump

    (e) Intracellular storage : Sacroplasmic Reticulum of muscle cells

14. Different ion channels have different ion selectivity
$\Rightarrow$ Voltage-gated Na^+ channel : Na^+ is 10 times permeable than K^+ (Na^+ is more permeable)
$\Rightarrow$ Voltage-gated K^+ channel : K^+ is 100 times permeable than Na^+ (K^+ is more permeable)

15. K^+ Channel Selectivity

   (a) K^+ can fully dehydrated because it can fully spaced to H_2O molecules (Energetically Favorable to shed its water shell)
   (b) K^+ has a larger size than Na^+ : Can interact stronger with carbonyl oxygen and fully shed its water shell
   (c) Na^+ distance is too far and solute is hard to remove
   (d) Dehydration allows ion enter the narrow selectivity filter of the channel
   $\Rightarrow$ Since Na^+ cannot fully dehydrated, entering is not favorable in K^+ channel

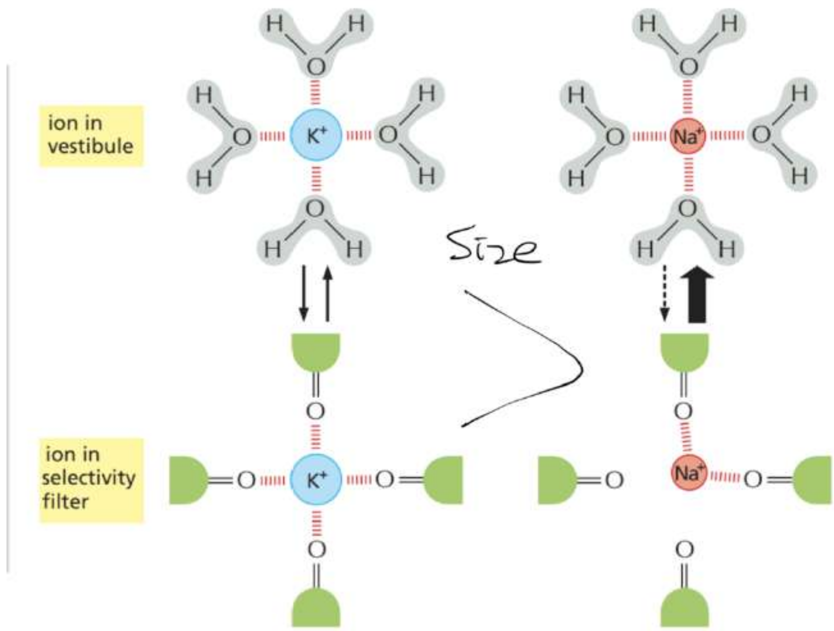

Figure 16: K^+ Channel Selectivity Reason

16. Gated ion channel respond stimuli

   - Voltage Gated
   - Ligand Gated (Extracellular Ligand) : Bind outside
   - Ligand Gated (Intracellular Ligand) : Bind inside
   - Mechanically Gated

17. Voltage-gated ion channel
    (a) Each channel composition : 4 subunits (K^+) or domains (Na^+, Ca^{2+})
    (b) Each domain composition : 6 transmembrane $\alpha-$ helices + 1 anti-parallel $\beta-$ sheet which lines the pore
    (c) Anti-parallel $\beta-$ sheet : Determine the ion selectivity of the channel
    (d) Open when the membrane is depolarized (Less negative)
    (e) Rate of Concentration = − (Rate of Voltage change)
    (f) Movement is started at resting potential = Equilibrium

Equilibrium potential is given by the Nernst equation

The Nernst equation is $$V = \frac{RT}{zF} \ln \frac{C_o}{C_i}$$

where

V = the equilibrium potential in volts (internal potential minus external potential)
C_o and C_i = outside and inside concentrations of the ion, respectively
R = the gas constant (8.3 J mol^{-1} K^{-1})
T = the absolute temperature (K)
F = Faraday's constant (9.6×10^4 J V^{-1} mol^{-1})
z = the valence (charge) of the ion
ln = logarithm to the base e

Figure 17: Resting Potential Equation

18. Steps for Voltage-Gated Ion Channel

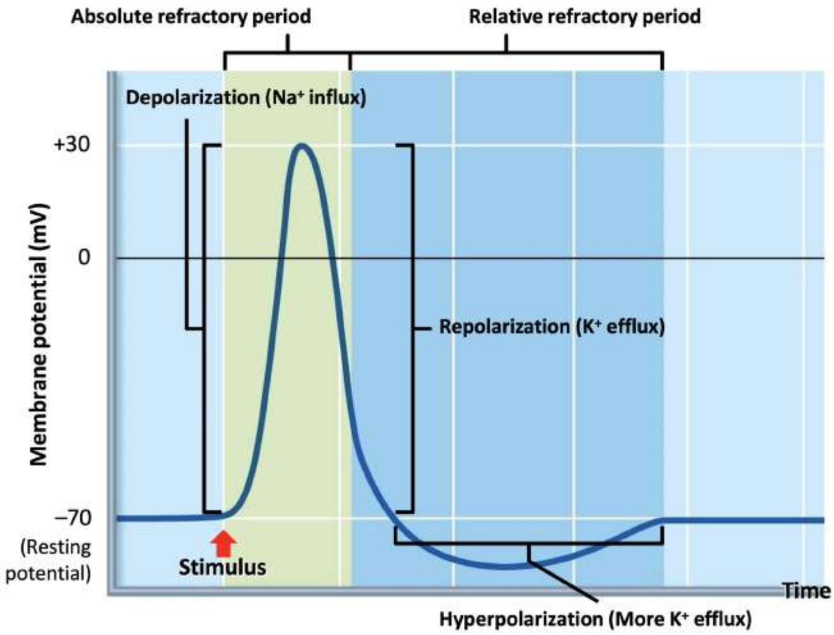

Figure 18: Voltage-Gated Ion Channel

(a) Resting Stage : -70 mV (Both Na^+ & K^+ are remain Closed state)

(b) Membrane potential reached to the threshold (-50 mV): Voltage-gated channel open

   i. Depolarization : -70 mV to +30 mV (Stimulus make Na^+ channel open

   ii. Na^+ channel is inactivated after brief opening

(c) Action Potential : Start decreasing from peak but still higher than resting potential

   - Early repolarization : Na^+ channels inactivated (not closed) but voltage gated K^+ channels open
   - Late repolarization : Closed to -70 mV
     $\Rightarrow$ Na^+ channels become closed state (More closed, Lower membrane potential)
   - Hyperpolarization : Lower than the resting potential
     $\Rightarrow$ Voltage gated K^+ channels remain open for the decreasing voltage
     $\Rightarrow$ All Na^+ channels is in closed state

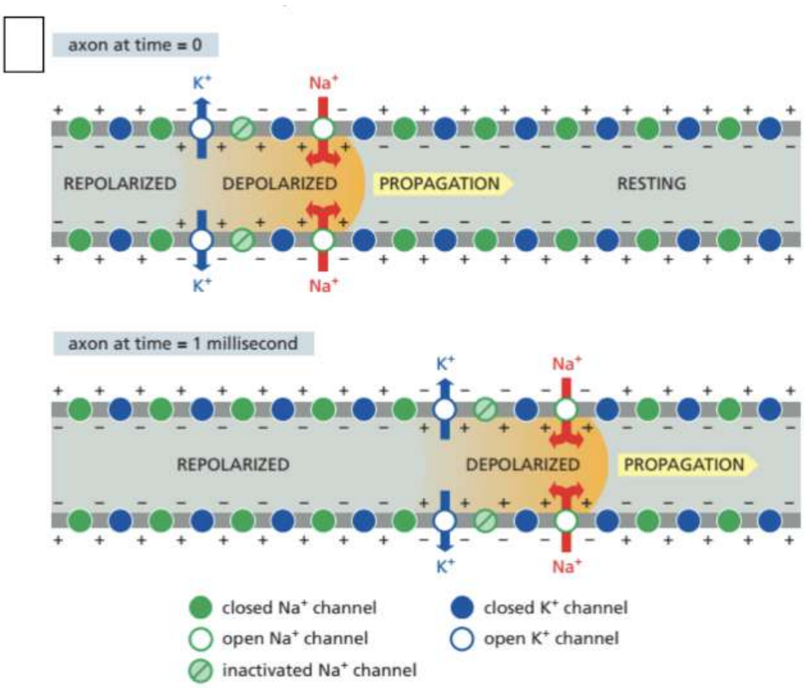

Figure 19: Irreversible Process

19. Nerve impulse : Directional (Cannot move backwards) = Voltage-gated channel Action Potential in neurons = The process is irreversible
    ⇒ Replorization cannot be changed or reversed

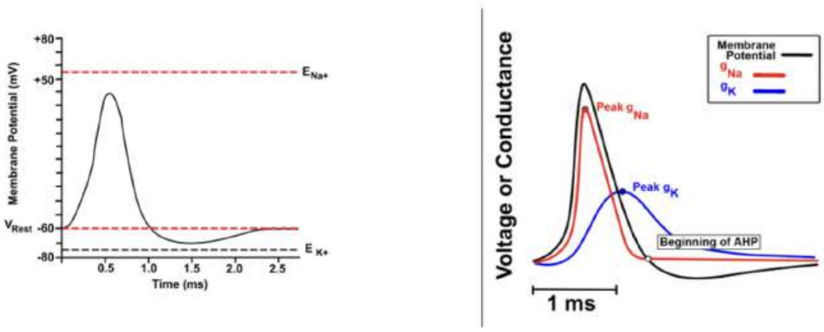

Figure 20: Irreversible Process

20. ***Myelination***
    - Surround by Glial cells

  - Glial cells in Central Nerve System (CNS) : Oligodendrocytes
  - Glial cells in Peripheral nervous system (PNS) : Schwann cells
- Glial cells : Form myelin sheath, cover the neuron and isolate the axonal membrane
  $\Rightarrow$ Stopnion leakage of the membrane potential
- Myelin Sheath stop at the node of Ranvier
- Node of Ranvier : High density of Na^+ channel
- Insulate neurons
- Increase the speed and efficiency of action potential propagation in nerve cells

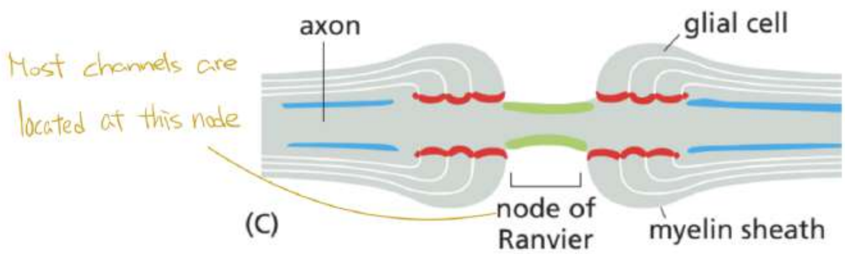

Figure 21: Myelination

21. Benefits of the action potential jumping node to node
    - Travel at a faster speed
    - Save metabolic energy
22. Transmitter-gated cation channels
    - Neurotransmitter bind to ligand-gated channels
    - Create a depolarization of membrane
    - High selective binding sites to respective neurotransmitter
    - Different ion selectivity in different channel
    - Insensitive to membrane potential (This does not meaning that must inhibit the formation of action potential; It could initiate this process too)

23. Neurotransmitter
    (a) Excitatory Neurotransmitter : Open cation channels and cause an influx of Na^+ ions $\rightarrow$ Used to depolarization and towards the threshold
    (b) Inhibitory Neurotransmitter : open anion channels and cause an influx of Cl^- ions $\rightarrow$ Used to hyperpolarization
24. Light-gated ion channels : **Channelrhodopsin**
    - From Algae
    - Responds to blue light (Very short)
    - Phototaxis (Favour lights)
    - Allows cations to pass
    - Have flagella : Allow them to swim towards the light
    - Application : Optogenetics : Control Localization by binding to a specialized protein

6 Lecture 11 Cytoskeleton & Junctions

1. Cytoskeleton : Protein filaments that extend throughout the cytoplasm
   ⇒ Dynamic network that help the cell change its shape, divided and respond to its environment
2. Types of protein filaments
   (a) Actin Filaments
       - Helical polymers of the protein actin
       - Flexible structure
       - Dispered throughout the cell
       - Highly concentrated in cortex : Below the plasma membrane
   (b) Intermediate Filaments
       - Ropelike fiber
       - E.g. Nuclear Lamina
       - Span the cytoplasm from one cell-cell junction to another
   (c) Microtubules
       - Long, holloe cylinder shape
       - Made of protein tubulin
       - Rigin than actin filaments
       - One end attached to a microtubule-organizing centre(MTOC) : Centosome
       - Function : Transport things to the cell
3. Connective Tissue : Have plenty ECM
4. Epithelial Tissue : Mainly composed of cells
5. Type of Junction in animals
   - Tight Junction : Seals gap between epithelial cells
   - Cell-Cell Anchoring Junctions : Adherens Junction + Desmosome

(a) Adherens Junction : Connect actin filaments in one cell within next cell
(b) Desmosome : Connect intermediate filaments in one cell to those in the next cell
(c) Form mechanical link between cells
(d) Between intermediate filaments and actin filaments between cells

- Channel-Forming Junction : Gap Junction

(a) Gap Junction : Allows small water-soluble molecules to transfer between cells
(b) Allow nutrients to pass through

- Cell-Matrix Anchoring Junction : Actin-linked Cell-matrix Junction + Hemidesmosome

(a) Actin-linked Cell-matrix Junction : Anchor actin filaments in cell to extracellular matrix
(b) Hemidesmosome : Anchor intermediate filaments in a cell to extracellular matrix

6. Tight Junction

- Form selective permeability barrier across epithelial cell sheet
- Form barriers to the movement of Water + Solutes + Cells from one body compartment to the others
- Maintain a directional transcellular transport of solutes from cell to cell

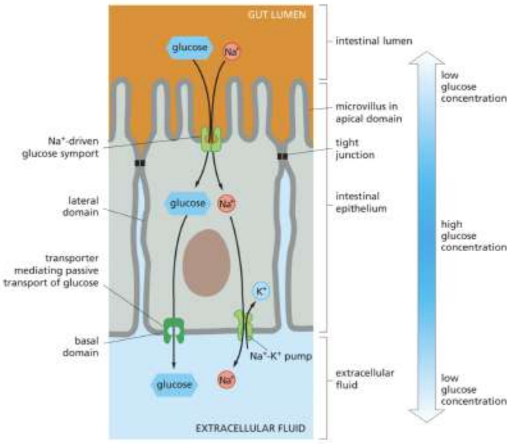

Figure 22: Glucose Transfer

- Tight-Junction Transmembrane Proteins : Claudin + Occludin
  (a) Claudin : Main components of the sealing strands
  $\Rightarrow$ Allows ions / small molecules to pass through
  (b) Occludin : Occurs in cell that do not have tight junctions
  $\Rightarrow$ Form barrier + Permeable to ions / small molecules
- Most things cannot pass through

7. Gap Junction

- Passive Junction (Can close, Not always open)
  (a) If something goes wrong, it can stop the spread and kill the whole epithelial tissue
  (b) Can isolate the wrong cell and kill it
- Allow small molecules to move from cell to cell
- Connected Metabolically + Electrically
- Connexins : Transmembrane Protein in gap junction
  $\Rightarrow$ Four putative membrane-spanning α helices
- Connexons : Six unit of connexins
  $\Rightarrow$ Contain Central Aqueous Pore
- Each gap junction is made by multi-units of connexins
- Regulated Permeability

(a) Speed of Opening and Closing : Rapid + Reversible
(b) Low Cytosolic pH : Closed
(c) $\uparrow$ Cytosolic Calcium Ion Level : Closed
$\Rightarrow$ Prevent the spreading of dangerous disturbance from cell to cell

8. Anchoring Junction

- Four types in vertebrates
(a) Adherens Junction (Cell to Cell)
(b) Desmosome (Cell to Cell)
(c) Actin-linked Cell Matrix Junction (Cell to Matrix)
(d) Hemidesmosome (Cell to Matrix)
- When forming anchoring junction
$\Rightarrow$ Need to cluster (Bounded) things together

9. Adherens Junctions and Cell Matrix Junction

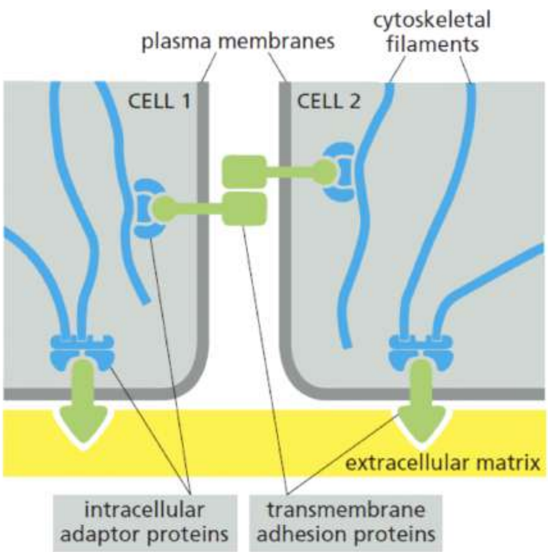

Figure 23: Adherens Junction

- Require intracellular adaptor proteins
- Adaptor bind with transmembrane adhesion protein

- External Linkage : Cell to Cell (Mediated by Cadherins) / Extracellular Matrix (Mediated by Integrins)
  ∴ Integrins are not Adherens Junction
- Internal Linkage : Indirect + Mediated by Intracellular Adaptor Protein

10. Adherens Junction (Epithelial Cells in the small intestine)

- Function 1 : Absorption of nutrition
- Microvilli (Finger-liked) : Increase the surface area
  ⇒ Absorb nutrients from the gut lumen
- Continuous Adhesion Belt : Tightly bound cells together

11. Cadherins

- Formed by glycoprotein
- Mediating Ca^{2+} depending binding
  ⇒ One Exception : Immunoglobulin-based
- Different cadherins locate at different place
  (a) Epithelial : E-cadherin
  (b) Neural : N-cadherin
  (c) Placenta : P-cadherin
- Have a Large Extracellular Segment + A Transmembrane Domain + An Intracellular Domain

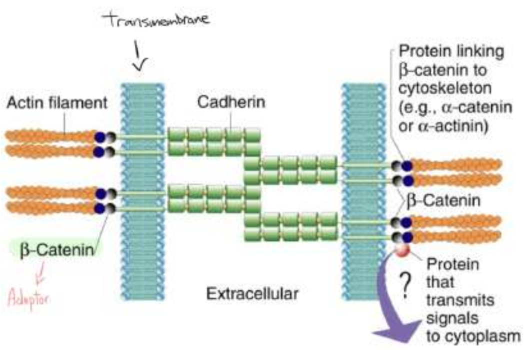

Figure 24: Cadherin

12. Cadherin-dependent Cell Sorting
    - Driven by **Cadherin-dependent adhesion**
    - Can recognize + Preferentially adhere to other cells of the same type
    - Non-random Process but guided by cadherin
    - Higher Expression, More sticky to each other
    - Example : Amphibian Embryo

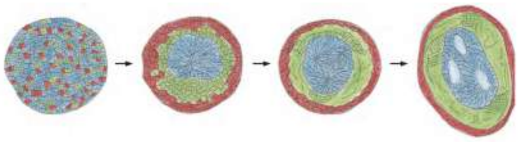

Figure 25: Cell Sorting Out

13. Desmosomes
    - Button-like Junction
    - Connect intermediate filaments of adjacent cell indirectly
    - Form continuous network throughout the tissue
    - Adaptor Protein + Transmembrane Protein
    - Type of intermediate filament (Intracellular Anchor)
      (a) Epithelial cell : Keratin Filaments
      (b) Heart Muscle cell : Desmin Filaments

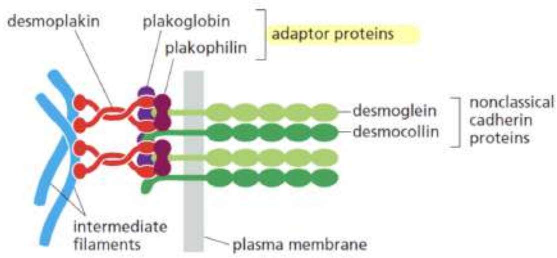

Figure 26: Desmosome

14. Hemidesmosomes

- Connect the basal surface of epithelial cells to underlying basal lamina
- Located at Epithelium & Connective Tissue
- Transmembrane Protein 1 : Integrins (Always in Pairs)
  ⇒ Example : $\alpha_6\beta_4$ Integrin (In Cell Membrane)
  ⇒ Inside Tail : Binds to the adapter proteins (Plectin / BP230)
  ⇒ Outside Head : Binds to laminin in the basal lamina
- Transmembrane Protein 2 : Collagen type XVII (Unusual)
  ⇒ Cooperate with the integrin, strengthening the adhesion complex
  ⇒ Tail & Head : Similar attachment to Integrins
- Intermediate Filament (Intracellular Anchor) : Keratin Filaments (Desmosome)
  ⇒ Provides tensile strength
- Adapted Protein : BP230
  ⇒ Creating a strong mechanical connection
- Laminin
  ⇒ Integrin grab in basal lamina

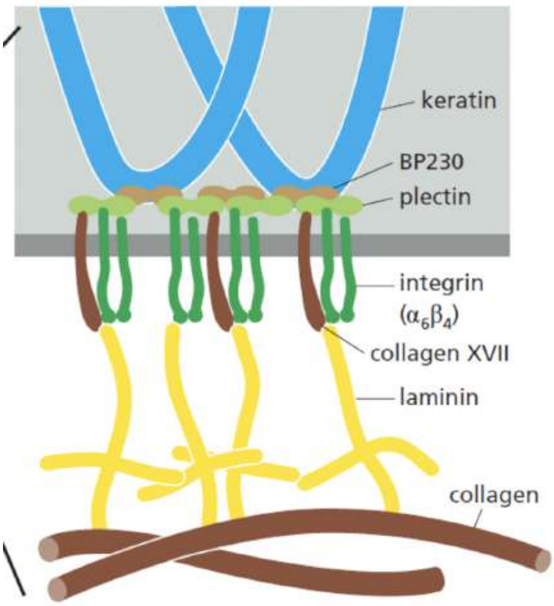

Figure 27: Hemidesmosomes

15. Actin-linked cell Matrix Junction : Integrins

- Composed of two transmembrane polypeptides (α and β)
- Both are non covalently linked
  $\Rightarrow$ Tail are Separated, but head is strong ligand binded
- Each has specific tissue localization
- Both chains have a small cytoplasmic domain (40-50 amino acid)
  $\Rightarrow$ Binds to components of the cytoskeleton
- The β chain has a RGD (Arg-Gly-Asp) binding site
  $\Rightarrow$ Bind to the RGD sequence on its ligands
- Appears at Extracellular Membrane Protein (Collagen, Fibronectin, Laminin) or Cell Surface Protein (ICAM, VCAM)
- α chain contains several Ca^{2+} binding sites
- When α (Stress Fibre) is active $\rightarrow$ α dissociate to β
  $\Rightarrow$ Link up with myosin and pull on the cell
- 1 β can bind with multiple α
- Kindlin : Bind to β subunit at strong adaptor-protein binding site
- Talin : Only bind when kindlin is binded to β subunit
  $\Rightarrow$ Allow actin filament to bind (Packed + Unfolded)
  $\Rightarrow$ IF adaptor protein bind to Talin, then have high affinity bind to ligand
  $\Rightarrow$ Vinculin : Bind between Talin and actin filament

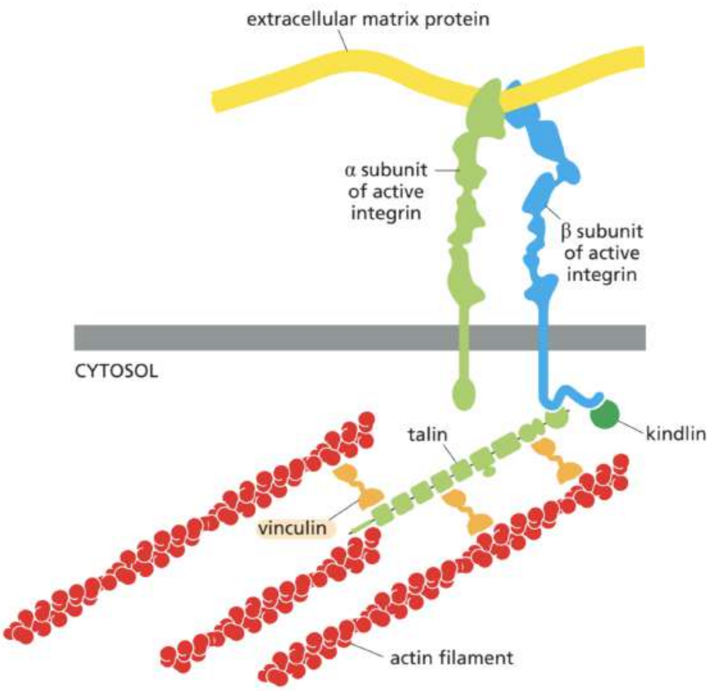

Figure 28: Integrin vs Actin Filaments

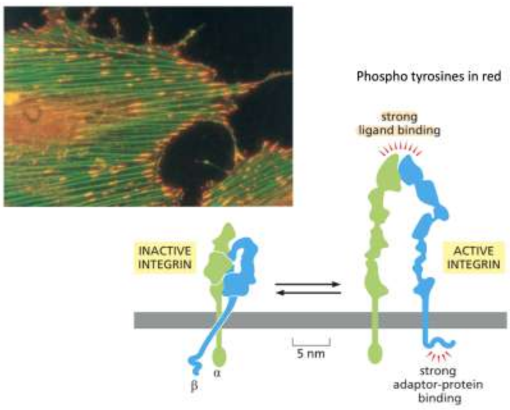

Figure 29: Inactive vs Active Integrins

16. Integrin Activation and Adhesion Formation

- Not always active (Inside-out Signaling)
- Active : Change its integrins from a low-affinity (Inactive, ”Closed”) state to a high-affinity (Active, ”open”) state for

binding

- Activation Pathway
  (a) An external signal binds to a receptor on the cell surface
  (b) The receptor triggers a signaling cascade inside the cell
  (c) Activates a small GTPase called Rap1
  ⇒ Can interacts with a protein called RIAM
  (d) Rap1 throw GDP and need GTP to interact with RIAM
  (e) RIAM : Bring regulator protein Talin to the plasma membrane→ Bringing it near the integrins
  (f) Talin is normally folded up on itself in the cytoplasm, which keeps it inactive
  (g) Talin head domain binds to a lipid called PIP_2 at the membrane, causing talin to be unfolded
  ⇒ This exposes its hidden binding sites
  (h) The unfolded Talin + Kindlin binds to the tail of the integrin β chain
  (i) This binding causes a conformational change in the integrin
  (j) The activated integrin can now strongly bind to its ligand in the extracellular matrix, forming adhesion

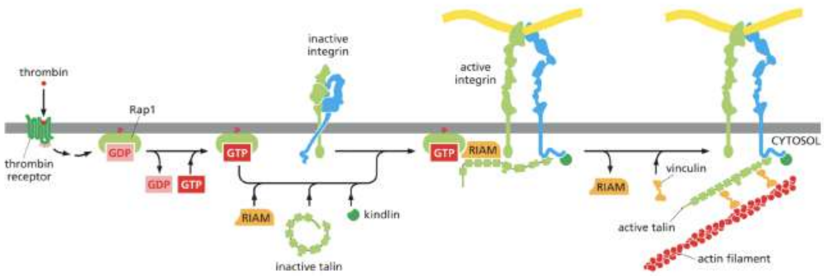

Figure 30: Intracellular Signaling

17. Selectins (Mediate transient actin-linked cell-cell junction adhesions)
    - Stronger than Cadherin (Adhesion Junction)

$\because$ Cadherin is mainly for transition interactions
$\Rightarrow$ E.g. Pull immune cell out of the bloodstream

- Mediating transient intercellular adhesion in the blood
- Type of Selectin
  (a) Endothelial Cells : E-selectin
  (b) Platelets : P-selectin
  (c) All kind of leukocytes : L-selectin
- Transmembrane Protein : Contain a large extracellular domain
  $\Rightarrow$ C-type lectin domain at N-terminus
- C-type Lectin Domain
  (a) Bind to a group of polysaccharides with 4 sugar residues on the end of carbohydrate chains on glycoproteins or glycolipids
  (b) Have 2 Ca^{2+} binding sites
  (c) Binding of selectins to sugars are Calcium-dependent

18. Immune System of Selectins

- Weak Adhesion & Rolling (Selectin-dependent) = Moving = Leukocyte Rolling
  (a) The white blood cell is flowing rapidly in the blood, briefly and weakly "sticks" to the inner wall of the blood vessel (the endothelial cells)
  (b) This causes it to slow down and roll along the vessel wall
  (c) Selectins expressed on the surface of the activated endothelial cells and white blood cells
  (d) They bind loosely and transiently to carbohydrate groups on the white blood cell
- Weak adhesion slows the white blood cell down from the fast-flowing blood, allowing it to sample the local environment for signals of infection or damage
- Strong Adhesion and Emigration (Integrin Dependent) = Stopping & Exiting

(a) When the white blood cell is rolling, it detects the activating signals from the damaged tissue
(b) The activating signals from the tissue switch the integrins to a high-affinity (Active) state
(c) Activated integrins then bind very strongly to their ligands on the endothelial cell surface
(d) White blood cell will stop rolling and stick on the vessel wall
(e) Then white blood cell squeezes (Emigrating) through the endothelial cell layer to enter the tissue

- Strong Adhesion and Emigration provides stable attachment for the cell to stop and then emigrate out of the blood vessel

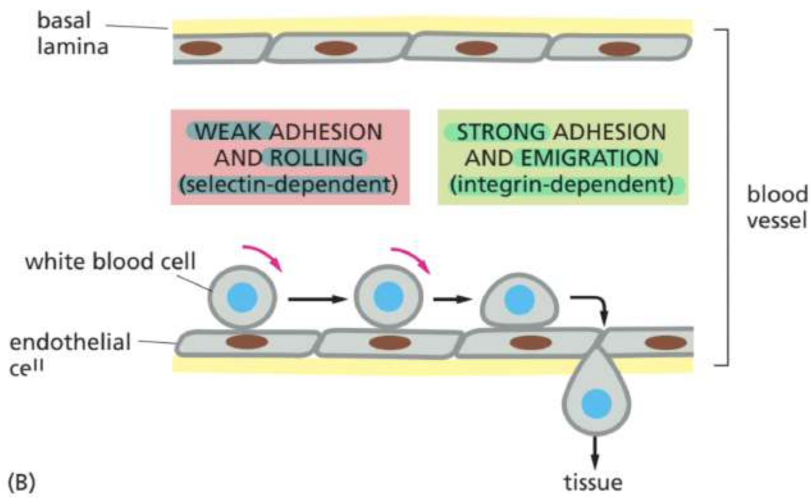

Figure 31: Selectins vs Integrins

19. Immunoglobulin Superfamily (IgSF)

- Immunoglobulin is composed of Ig domains
  ⇒ A tightly folded structure
- Not only found in immunoglobulin, but also in other protein
  ⇒ The whole superfamily : IgSF
- Most of the proteins in IgSF are membrane proteins on the

surface of lymphocytes that involve in various immune functions

- Some of these proteins mediate calcium-independent cell-cell adhesion
  $\Rightarrow$ E.g. N-CAM (Neural cell adhesion molecule), ICAMs (Intercellular adhesion molecules)
- The extracellular domain of N-CAM binds to the N-CAM of adjacent cells
- ICAM binds to integrin on adjacent cells
- Different cell adhesion molecules
  (a) Homophilic : Involve binding between similar molecules
  $\Rightarrow$ Most of them are in this group
  $\Rightarrow$ E.g. Cadherins (Calcium Dependence), Nectins (Calcium Independent)
  (b) Heterophilic (w/integrins) : Involve binding between different molecules
  $\Rightarrow$ E.g. Integrins & Selectins (Calcium Dependence)
  (c) Homo/Hetero (with Nectins): Can do both
  $\Rightarrow$ E.g. Nectins & IgSF Proteins (Calcium Independent)

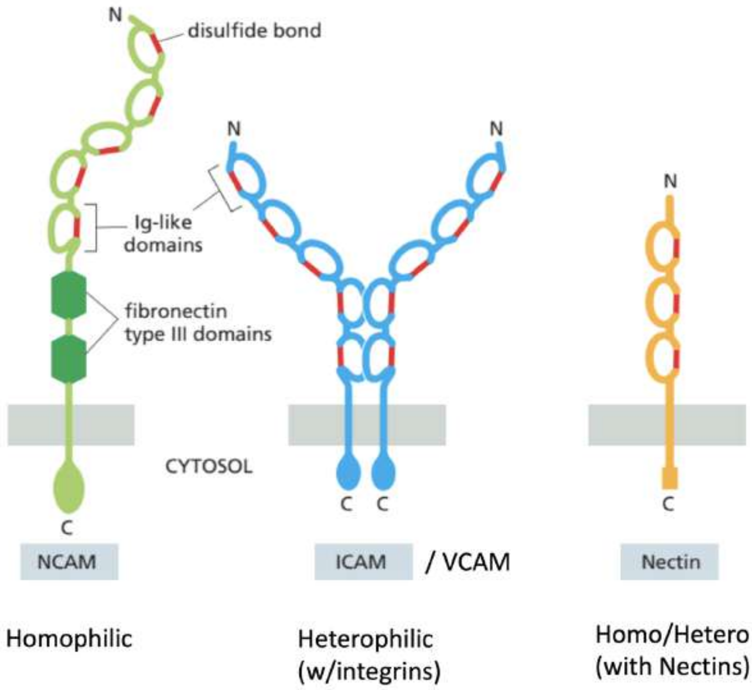

Figure 32: Different cell adhesion molecules